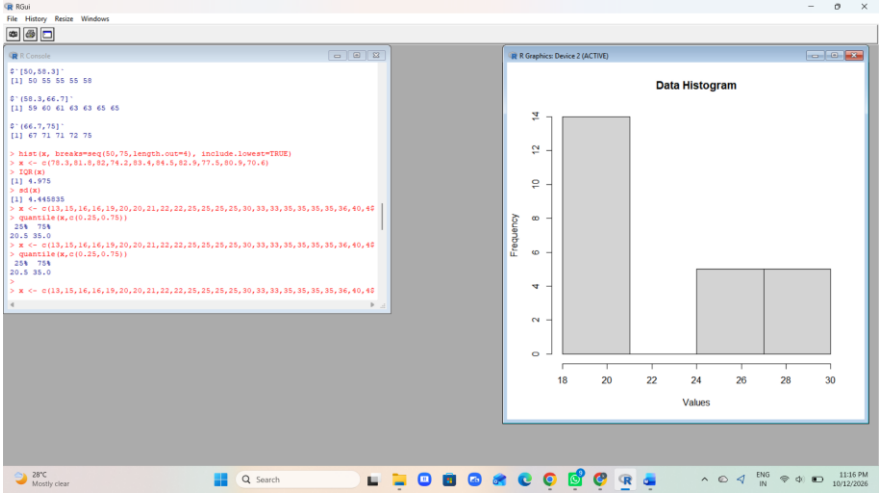
Experiment-1

```
> x <- matrix(c(18,22,20,2,28,40,20,10,40), nrow=3, byrow=TRUE)
> colnames(x) <- c("A","B","C")
> rownames(x) <- c("5-6","7-8","9-10")
>
> cov(x[,2], x[,3])
[1] -20
> cov(x)
           A   B         C
A  97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
> cor(x[,2], x[,3])
[1] -0.1889822
> cor(x)
           A          B          C
A  1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
>
```

Experiment-2


RGui
File History Resize Windows
R Console
$`[50,58.3]`
[1] 50 55 55 55 58
$`(58.3,66.7]`
[1] 59 60 61 63 63 65 65
$`(66.7,75]`
[1] 67 71 71 72 75
> hist(x, breaks=seq(50,75,length.out=4), include.lowest=TRUE)
> x <- c(78.3,81.8,82,74.2,83.4,84.5,82.9,77.5,80.9,70.6)
> IQR(x)
[1] 4.975
> sd(x)
[1] 4.465835
> x <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45
> quantile(x,c(0.25,0.75))
25% 75%
20.5 35.0
> x <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45
> quantile(x,c(0.25,0.75))
25% 75%
20.5 35.0
>
> x <- c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45
R Graphics: Device 2 (ACTIVE)
Data Histogram
Frequency
0
2
4
6
8
10
12
14
18
20
22
24
26
28
30
Values
28°C
Mostly clear
Search
ENG
IN
11:16 PM
10/12/2026

Experiment-3

```
> x <- c(1,1,5,5,5,5,5,8,8,10,10,10,10,12,14,14,14,15,15,15,15,15,15,18,18,18,18,18)
>
> b <- split(x, rep(1:3, length.out=length(x)))
> b
$`1`
 [1]  1  5  5 10 10 14 15 15 18 18

$`2`
[1]  1  5  8 10 12 14 15 15 18

$`3`
[1]  5  5  8 10 14 15 15 18 18

> lapply(b, function(z) rep(mean(z), length(z)))
$`1`
 [1] 11.1 11.1 11.1 11.1 11.1 11.1 11.1 11.1 11.1 11.1

$`2`
[1] 10.88889 10.88889 10.88889 10.88889 10.88889 10.88889 10.88889 10.88889
[9] 10.88889

$`3`
[1] 12 12 12 12 12 12 12 12 12

> lapply(b, function(z) rep(median(z), length(z)))
$`1`
 [1] 12 12 12 12 12 12 12 12 12 12

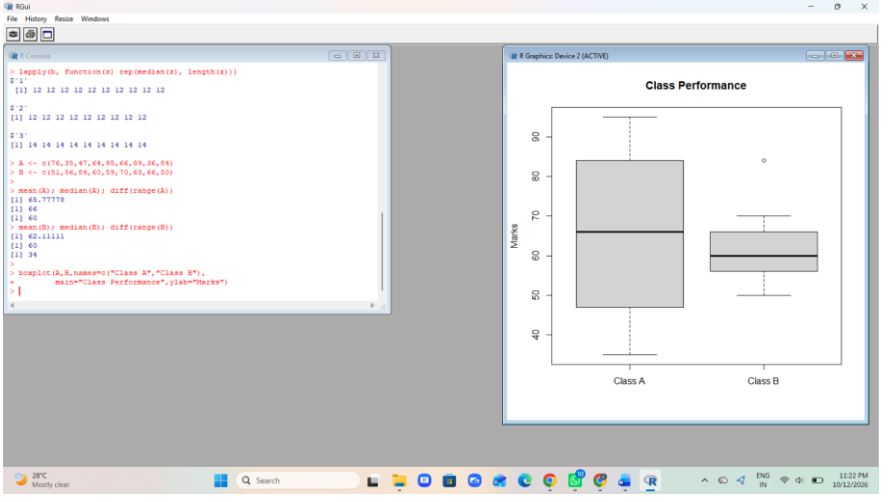
$`2`
[1] 12 12 12 12 12 12 12 12 12

$`3`
[1] 14 14 14 14 14 14 14 14 14

>
```



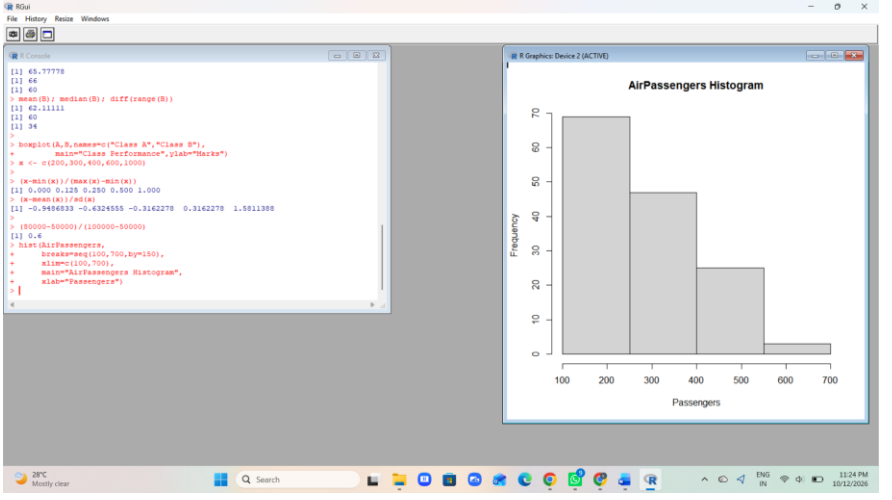
Experiment-4



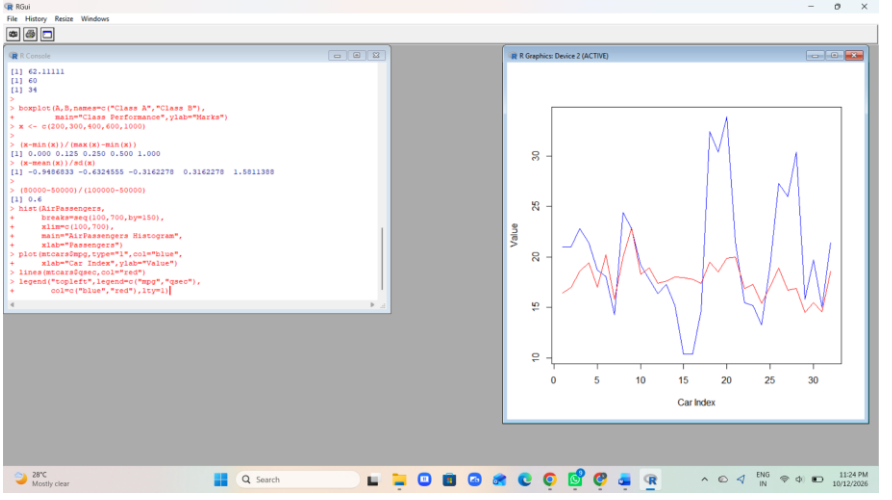
Experiment-5

```
+          main="Class Performance",ylab="Marks")
> x <- c(200,300,400,600,1000)
>
> (x-min(x))/(max(x)-min(x))
[1] 0.000 0.125 0.250 0.500 1.000
> (x-mean(x))/sd(x)
[1] -0.9486833 -0.6324555 -0.3162278  0.3162278  1.5811388
>
> (80000-50000)/(100000-50000)
[1] 0.6
>
```

Experiment-6



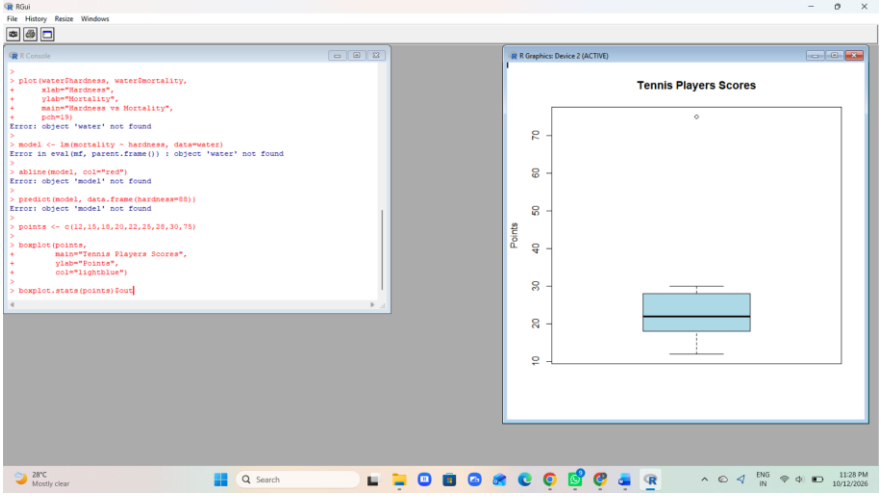
Experiment -7



Experiment 8:

```
>
> predict(model,data.frame(hardness=88))
Error: object 'model' not found
>
> data(water)
Warning message:
In data(water) : data set ‘water’ not found
>
> plot(water$hardness, water$mortality,
+      xlab="Hardness",
+      ylab="Mortality",
+      main="Hardness vs Mortality",
+      pch=19)
Error: object 'water' not found
>
> model <- lm(mortality ~ hardness, data=water)
Error in eval(mf, parent.frame()) : object 'water' not found
>
> abline(model, col="red")
Error: object 'model' not found
>
> predict(model, data.frame(hardness=88))
Error: object 'model' not found
>
>
```

Experiment-9:


RGui
File History Resize Windows
R Console
> plot(water$hardness, water$mortality,
+ xlab="Hardness",
+ ylab="Mortality",
+ main="Hardness vs Mortality",
+ pch=19)
Error: object 'water' not found
> model <- lm(mortality ~ hardness, data=water)
Error in eval(mf, parent.frame()) : object 'water' not found
> abline(model, col="red")
Error: object 'model' not found
> predict(model, data.frame(hardness=88))
Error: object 'model' not found
> points <- c(12,15,18,20,22,25,28,30,75)
> boxplot(points,
+ main="Tennis Players Scores",
+ ylab="Points",
+ col="lightblue")
> boxplot.stats(points)$out
R Graphics: Device 2 (ACTIVE)
Tennis Players Scores
Points
10
20
30
40
50
60
70
28°C
Mostly clear
Search
ENG
IN
11:28 PM
10/12/2026

Experiment 10

